MILITARY SPECIFICATION

LEAD, EXPLOSIVE

MARK 28 MOD 0

MARK 29 MOD 0

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of two types of explosive leads, as classified in 1.2 and described in 3.1.

1.2 Classification. Leads covered by this specification are of the following types:

Type I Lead, Explosive, MARK 28 MOD 0	Drawing 1137AS2161.
Type II Lead, Explosive, MARK 29 MOD 0	Drawing 1137AS2164.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1336

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SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for.
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice, Process for).
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.
MIL-C-21723	Composition CH-6.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents and drawings. The following other Government documents and drawings of the issue specified in the solicitation form a part of this specification to the extent specified herein.

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OTHER GOVERNMENT DOCUMENTS

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

DEPARTMENT OF DEFENSE (DoD)

DoD 4145.26M	DoD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (Code Ident 30003)

1137AS2161	Lead, Explosive, MARK 28 MOD 0.
1137AS2164	Lead, Explosive, MARK 29 MOD 0.
SA1137AS2167	Fixture, Vibration Test.
SA1137AS2168	Fixture, Jolt Test.
SA1137AS2169	Fixture, 12-Meter Drop Test.
1137AS2171	Container, Shipping and Storage (for 500 MARK 28 MOD 0 Explosive Leads) Packing and Marking.
1137AS2172	Container, Shipping and Storage (for 100 MARK 29 MOD 0 Explosive Leads) Packing and Marking.

(Copies of drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A-108	Steel Bars, Carbon, Cold-finished, Standard Quality, Specification for.
ASTM A-582	Free-machining Stainless and Heat-resisting Steel Bars, Hot-rolled or Cold-finished, Specification for.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Item description. The Type II lead performs a transfer function to detonate a Type I lead in a safety-arming (S-A) device.

3.1.1 Government-furnished property. Unless otherwise specified in the contract or purchase order, the Government will furnish the contractor with the following components in a sufficient quantity to perform the function and output test specified herein (see 4.6.2 and 6.2.3):

a. MARK 9 MOD 0 Explosive Leads in accordance with LD 480141.

b. Exploding bridgewire (EBW) detonators in accordance with 4.6.2.1.2.a.

3.2 Characteristics. Unless otherwise specified herein, the following requirements apply for both the Type I lead and the Type II lead.

3.2.1 Explosive charge moisture content. The explosive charge in the lead shall be Composition CH-6 in accordance with MIL-C-21723. The moisture content of the explosive charge shall not exceed 0.20 percent at the time of loading (see 4.6.1).

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3.2.2 Performance.

3.2.2.1 Function and output. The lead shall function on the first and only try when initiated in accordance with 4.6.2. The explosive output shall meet the acceptance criteria specified in 4.2.2 for the first article and in 4.3.1.3 for the inspection lot.

3.2.2.2 Environmental. The lead shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the following environments.

3.2.2.2.1 Transportation vibration-temperature. The lead shall meet the requirements of 3.2.2.1 after exposure to the environment of 4.6.3.1.

3.2.2.2.2 Temperature and humidity. The lead shall meet the requirements of 3.2.2.1 after exposure to the environment of 4.6.3.2.

3.2.2.2.3 Flight vibration. The lead shall meet the requirements of 3.2.2.1 after exposure to the environment of 4.6.3.3.

3.2.2.3 Safety.

3.2.2.3.1 Jolt. The lead shall be safe to handle and dispose of after exposure to the environment of 4.6.4.1.

3.2.2.3.2 Twelve-meter drop. The lead shall be safe to handle and dispose of after exposure to the environment of 4.6.4.2.

3.2.3 Radiographic. The lead shall be free from defects such as cracks, voids, fissures, or loose powder when radiographically inspected (see 4.6.5).

3.2.4 Visual. There shall be no evidence of external damage, such as cracks, dents, pits, burrs, or other blemishes, when visually inspected (see 4.6.6).

3.3 First article. When specified in the contract or purchase order, a sample of Type I or Type II leads shall be subjected to first article inspections (see 4.2 and 6.2.4).

3.4 Design and construction.

3.4.1 Production drawings. The lead shall meet all the requirements of and shall be manufactured and assembled in accordance with Drawing 1137AS2161 and all associated documents for Type I and on Drawing 1137AS2164 and all associated documents for Type II.

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3.4.2 Safety precautions. As a minimum, the safety precautions of DoD 4145.26M shall apply during manufacture, handling, and inspection of the leads, but shall not be considered as a substitution for the contractor's own safety procedures developed to accommodate the hazards associated with the lead.

3.4.3 Standards of manufacture.

3.4.3.1 Product cleanliness. No oils or solid lubricants of any kind shall be used on any part of the lead after final cleaning of components, except as specified on the drawings.

3.4.3.2 Working environment. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

3.4.3.3 Workmanship. Workmanship shall meet the requirements of MIL-A-2550.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

4.2 First article inspections. When specified in the contract or purchase order, first article inspections shall be performed by the contractor after award of contract and prior to production (see 6.2.4). As specified in the contract or purchase order, first article inspections shall be performed on 50 Type I or 50 Type II leads which have been produced with the equipment, material, and procedures to be used in production (see 6.2.4). First article approval is valid only on the contract under which it is granted unless extended by the Government to other contracts. Any production prior to approval of the first article shall be at the risk of the contractor.

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4.2.1 First article inspection sequence. The first article of 50 Type I or 50 Type II leads shall be subjected to the inspections of Figure 1 in the sequence shown. At the conclusion of the inspection sequence and after receipt of first article approval, any remains of the first article shall be disposed of at the discretion of the cognizant demolition officer.

4.2.2 First article acceptance and rejection criteria. Each first article lead shall meet the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 1, the applicable requirements of Section 3 shall be met on the first and only trial; one or more failures shall be cause for refusal to grant first article approval. For the function and output test of 4.6.2, acceptance or rejection of the first article shall be based on the following criteria:

a. A depth of dent of less than 0.075 inch from any Type I sample lead or less than 0.038 inch from any Type II sample lead, when measured in accordance with 4.6.2.3.c, shall be cause for rejection of the first article.

b. The first article shall be accepted if the quantity $(\bar{X}-L)/s$, when computed in accordance with 4.6.2.3.d, is equal to or greater than the k value of MIL-STD-414, Table B-1, Acceptable Quality Level (AQL) of 0.65 percent defective. A quantity less than the k value shall be cause for rejection of the first article.

4.3 Quality conformance inspections.

4.3.1 Inspection lot. As specified in the contract or purchase order, an inspection lot shall consist of completed Type I or Type II leads, fabricated under the same conditions and offered for acceptance inspection at the same time (see 6.2.1).

4.3.1.1 Inspection lot size. The inspection lot size shall be as specified in the contract or purchase order (see 6.2.1).

4.3.1.2 Inspection lot acceptance inspection sequence. After the inspection lot has been formed and identified by lot number, the lot shall be subjected to the inspections of Figure 2 in the sequence shown. Where specified during the inspection sequence, a quantity of leads shall be randomly sampled from the inspection lot in accordance with MIL-STD-414, Inspection Level IV, Standard Deviation Method, Variability Unknown. As the sample continues through the inspection sequence, the remainder of the lot shall be held in bond. If the sample does not cause lot rejection (see 4.3.1.3), the remainder of the lot shall continue through the inspection sequence as shown on Figure 2. Upon completion of the inspection sequence, the remains of the sample leads shall be disposed of at the discretion of the cognizant demolition officer.

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4.3.1.3 Inspection lot acceptance and rejection criteria. Each lead of the inspection lot shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 2, each sample lead shall meet the applicable requirements of Section 3 on the first and only trial. For the function and output test of 4.6.2, acceptance or rejection of the inspection lot shall be based on the following criteria:

a. A depth of dent of less than 0.075 inch from any Type I sample lead or less than 0.038 inch from any Type II sample lead, when measured in accordance with 4.6.2.3.c, shall be cause for rejection of the inspection lot.

b. The inspection lot shall be accepted if the quantity ($\overline{X}$-L/s, when computed in accordance with 4.6.2.3.d, is equal to or greater than the k value of MIL-STD-414, Table B-1, AQL of 0.65 percent defective. A quantity less than the k value shall be cause for rejection of the inspection lot.

4.4 Measuring and test equipment.

4.4.1 Calibration and maintenance. All measuring and test equipment utilized by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662. When specified in the contract or purchase order, the contractor shall prepare a Calibration System Description in accordance with MIL-STD-45662 (see 6.2.2). The Calibration System Description shall be accepted by the Government prior to use.

4.4.2 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements, under the conditions, and to the accuracies specified herein. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured.

4.4.3 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein, are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.4.1 and 4.4.2, and are subject to approval by the contracting activity.

4.4.4 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. When specified in the contract or

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purchase order, the contracting activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.5 Test conditions and provisions. The following test conditions and provisions shall apply during performance of the inspection methods of 4.6:

a. Test environment. Unless otherwise specified herein, the test environment shall conform to 3.4.3.2.

b. Test fixtures. Unless otherwise specified in the contract or purchase order, all test fixtures required for performance of the inspections of 4.6, other than those specified herein, shall be configured and fabricated by the contractor and shall be approved by the contracting activity prior to use (see 6.2.1).

c. Test plan. When specified in the contract or purchase order, the contractor shall prepare a test plan for the inspections specified herein (see 6.2.2). The test plan shall be prepared in accordance with MIL-T-18303 and shall be accepted by the Government prior to use. The contractor shall not deviate from the accepted test plan unless authorized by the Government.

d. Inspection sequences. The inspections of 4.6 shall be performed in the sequences shown on Figure 1 or 2, as applicable. When feasible, sequential inspections may be combined provided it can be demonstrated by the contractor that the applicable requirements of Sections 3 and 5 have been individually met. The contractor shall not deviate from the inspection methods of 4.6 and additional tests shall not be added unless authorized by the Government.

4.6 Inspection methods. To determine conformance to the requirements of Sections 3 and 5, the following inspections shall be performed in accordance with the test conditions and provisions of 4.5.

4.6.1 Explosive charge moisture content. To determine conformance to 3.2.1, the moisture content of the explosive charge shall be measured as specified herein on each day of production or on each lot, whichever requires the more frequent testing, plus such additional tests as the cognizant inspector considers warranted by variations in humidity, manufacturing process or materials. An approximate 2.0-gram sample of the

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material under investigation shall be collected in a tared weighing bottle, having a cap or stopper ground to fit, at the site of and during the production of the lot. The stoppered bottle and contents shall be accurately weighed. After weighing, the stopper shall be removed, and the open bottle and contents shall be dried at 212 ±3°F for a period of 45 ±5 minutes, cooled in a desiccator, stoppered, and reweighed on the same balance that was used for the first measurement. The loss in weight of the stoppered bottle and contents shall be calculated as the percentage of moisture in the sample. Nonconforming compositions shall be rejected and discarded.

WARNING

The lead contains explosive material which is sensitive and hazardous. Suitable safety precautions shall be observed during handling and inspection of the lead (see 3.4.2).

4.6.2 Function and output. The function and output test consists of initiating the lead on a steel dent block (see 4.6.2.1.1.e) and measuring the depth of the dent to determine the output. The test operation and selection of test equipment and materials, except for the test equipment and materials listed in 4.6.2.1.1, shall be the responsibility of the contractor. A list of recommended test equipment and materials is given in 4.6.2.1.2 to facilitate test setup by the contractor. To determine conformance to 3.2.2.1, test the lead in accordance with 4.6.2.2; then measure the depth of the dent and perform the computation as specified in 4.6.2.3.

4.6.2.1 Function and output test equipment and materials.

4.6.2.1.1 Required test equipment and materials. The following test equipment and materials are required to perform the function and output test:

a. Lead and detonator holder, spacer, and sleeve in accordance with Figure 3 when testing the Type I lead.

b. Lead and detonator holder in accordance with Figure 4 when testing the Type II lead.

c. MARK 9 MOD 0 Explosive Lead when testing the Type I lead (see 3.1.1).

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d. Detonator (an electric EBW detonator is recommended; see 3.1.1 and 4.6.2.1.2.a).

NOTE: Unless otherwise specified in the contract or purchase order, the Government will supply the contractor with EBW detonators (see 3.1.1). A contractor-selected detonator may be used subject to the approval of the output characteristics by the contracting activity. Approval of the contractor-selected detonator does not imply approval of the detonator's safety characteristics.

e. Dent block in accordance with Figure 5.

f. Support blocks to support the dent block of e. above during the measurement procedure of 4.6.2.3.

g. Dial indicator height gage and probe as specified in MIL-STD-331, Test 301.

h. Detonator firing source, low current resistance measurement device with the current limited to 10 milliamperes (mA) maximum, and interconnecting cables for the detonator of d. above.

4.6.2.1.2 Recommended test equipment and materials. The following test equipment and materials are recommended for safe performance of the function and output test in conjunction with suitable safety procedures and facilities (see 3.4.2):

a. EBW detonator of 0.190 ±0.001 inch diameter with an output charge of 70 ±10 milligrams of pentaerythritol tetranitrate (PETN) pressed at 10,000 ±500 pounds per square inch (lb/in^2) (see 3.1.1).

b. Appropriate firing bay or barricades for containing the blast and fragment hazards of the lead.

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4.6.2.2 Function and output test method. Test the lead as specified below:

WARNING

The function and output test involves hazardous operations and requires the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 3.4.2).

a. Setup the test assembly as shown on Figure 6 for the Type I lead and as shown on Figure 7 for the Type II lead, except, in both cases, do not install the detonator.

NOTE 1: The installation and wiring of the detonator shall be performed in accordance with the contractor's safety procedures (see Step f. below).

NOTE 2: Steps b. through j. below are suggested as good safety procedures appropriate for the hazards involved.

b. Clear the firing area of all nonessential personnel and equipment.

c. Verify that the detonator leads are shorted and that ordnance personnel, the firing cables, and the lead test assembly are grounded.

d. Verify that the firing cables are not connected to the firing source.

e. Insert the detonator into a hand-safe containment block, remove the short from the detonator leads, connect the detonator leads to the firing cable, and insulate the connections. (NOTE: A hand-safe containment block is a receptacle which provides adequate protection for safe handling of an explosive item by personnel.)

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f. After removing the detonator from the hand-safe containment block, insert the detonator into the lead and detonator holder as shown on Figure 6 for Type I and on Figure 7 for Type II so the detonator is bottomed against the lead in the holder.

g. Remove all remaining personnel and nonexpendable equipment from the test bay and verify that all openings through which fragments could escape are secured.

WARNING

The lead causes a significant blast and fragment hazard. Verify that the area is cleared of all personnel before applying energy to the firing circuit (see 3.4.2).

h. Connect the firing cable to the firing source.

i. Verify that all personnel are clear, sound the firing alarm, and initiate the detonator.

j. Allow sufficient time to elapse for the fumes to clear the firing bay, resecure the firing source, disconnect and short the firing cables, and retrieve the dent block.

4.6.2.3 Output dent measurement procedure. Measure the depth of dent as follows:

a. Remove all foreign material from the dent block. Place the dent block on the support blocks (see 4.6.2.1.1.f) as shown on Figure 5.

b. As shown on Figure 5, place the gage probe of the dial-indicator height gage (see 4.6.2.1.1.g) in the deepest part of the dent. Zero the gage indicator.

NOTE: Do not change the relative position of the dent block on the support blocks after zeroing the gage indicator.

c. Remove the probe from the dent and take gage readings at two points, each approximately 0.125 ±0.030 inch from the opposite edge of the dent block as shown on Figure 5. Average the two readings to determine the depth of dent (see 4.2.2 and 4.3.1.3).

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d. Using the data from this test, compute the quantity $(\overline{X}-L)/s$, as defined in MIL-STD-414, Single Specification Limit, Form 1, Variability Unknown, Standard Deviation Method. The lower limit, L, of the booster output to be used in the computation shall be 0.085 inch for Type I and 0.041 inch for Type II (see 4.2.2 and 4.3.1.3).

4.6.3 Environmental.

4.6.3.1 Transportation vibration-temperature. To determine conformance to 3.2.2.2.1, insert the lead into the test fixture of Drawing SA1137AS2167. Test the lead in accordance with MIL-STD-331, Test 119, Procedure 1.

4.6.3.2 Temperature and humidity. To determine conformance to 3.2.2.2.2, test the lead in accordance with MIL-STD-331, Test 105, Single Chamber-Cycling Method, except the test shall be conducted for one 14-day cycle.

4.6.3.3 Flight vibration. To determine conformance to 3.2.2.2.3, insert the lead into the test fixture of Drawing SA1137AS2167. Test the lead in accordance with MIL-STD-810, Method 514.2, Procedure IIA, except the vibration spectrum shall be in accordance with Figures 8 and 9. For each of the three mutually perpendicular axes, the vibration spectrum of Figure 8 shall be applied for 60 minutes minimum followed by vibration in accordance with Figure 9 for 5 minutes minimum.

4.6.4 Safety.

4.6.4.1 Jolt. To determine conformance to 3.2.2.3.1, insert the lead into the test fixture of SA1137AS2168. Test the lead in accordance with MIL-STD-331, Test 101, Procedure II.

4.6.4.2 Twelve-meter drop. To determine conformance to 3.2.2.3.2, insert the 10 first article leads into the test fixture of Drawing SA1137AS2169. Test the lead in accordance with MIL-STD-331, Test 103, Procedure I.

NOTE: The fixture of Drawing SA1137AS2169 is designed to accommodate the five impact orientations required by MIL-STD-331 in one drop, at two leads per orientation.

4.6.5 Radiographic. To determine conformance to 3.2.3, perform a radiographic inspection in accordance with MIL-STD-453, Quality Level 2. Take a minimum of two X-rays of each lead as follows: Take the first X-ray showing the longitudinal section through the lead (see Drawing 1137AS2161 for Type I or Drawing 1137AS2164 for Type II).

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Rotate the lead 90 ±5 degrees about the longitudinal axis and then take the second X-ray, again showing the longitudinal section through the lead.

4.6.6 Visual. To determine conformance to 3.2.4, visually inspect the lead.

4.6.7 Packaging. To determine conformance to Section 5, inspect the packaging, packing, and marking of the lead prior to shipment.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C, and as specified in 5.1.1.

5.1.1 Level C packaging. Leads shall be packaged in accordance with Drawing 1137AS2171 for Type I or Drawing 1137AS2172 for Type II, and in a manner which provides adequate safety and protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity. Packaging shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and as specified in 5.2.1.

5.2.1 Level C packing. Leads, packaged in accordance with 5.1, shall be packed in accordance with Drawing 1137AS2171 for Type I or Drawing 1137AS2172 for Type II, and in a manner which provides adequate safety and protection from damage during direct shipment from the supply source to the first receiving activity. Packing and containers shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with Drawing 1137AS2171 for Type I leads or Drawing 1137AS2172 for Type II leads, and MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The Type I lead and the Type II lead are intended for use in the AIM-54C Phoenix Guided Missile.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsiblity for inspection and inspection facilities, if other than as specified in 4.1.

c. Acquisition of Type I or Type II leads (see 4.3.1).

d. Inspection lot size (see 4.3.1.1).

e. Government final inspection gage drawings, if required (see 4.4.4).

f. Approval of test fixtures by contracting activity (see 4.5.b).

g. Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulations (DAR) 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirements	Applicable DID	Options
4.4.1	Calibration System Description	DI-R-7064	---
4.5.c	Test Plan	DI-T-5204	---

(DIDs related to this specification and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, Acquisition Management Systems and Data Requirements Control List (AMSDL). Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

6.2.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of either 50 Type I leads or 50 Type II leads, as specified in the contract or purchase order (see 3.3 and 4.2). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Preparing activity:
Navy - AS
(Project 1336- N475

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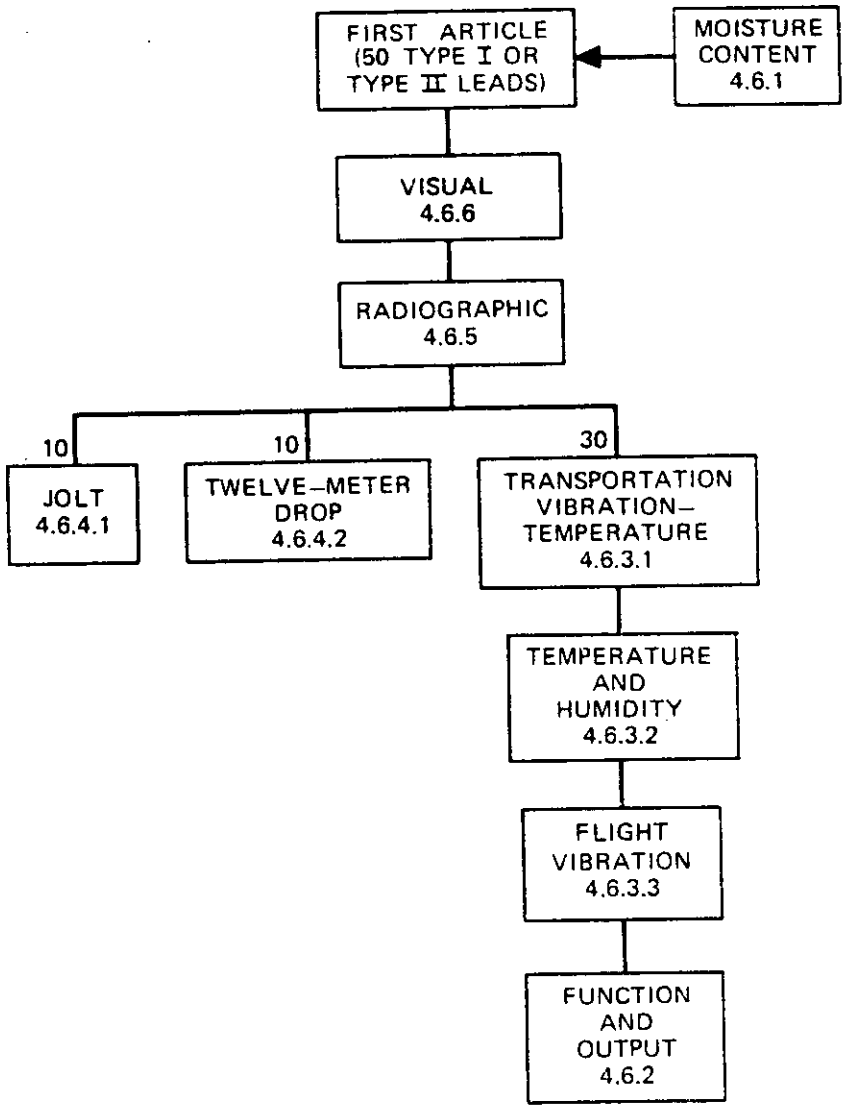
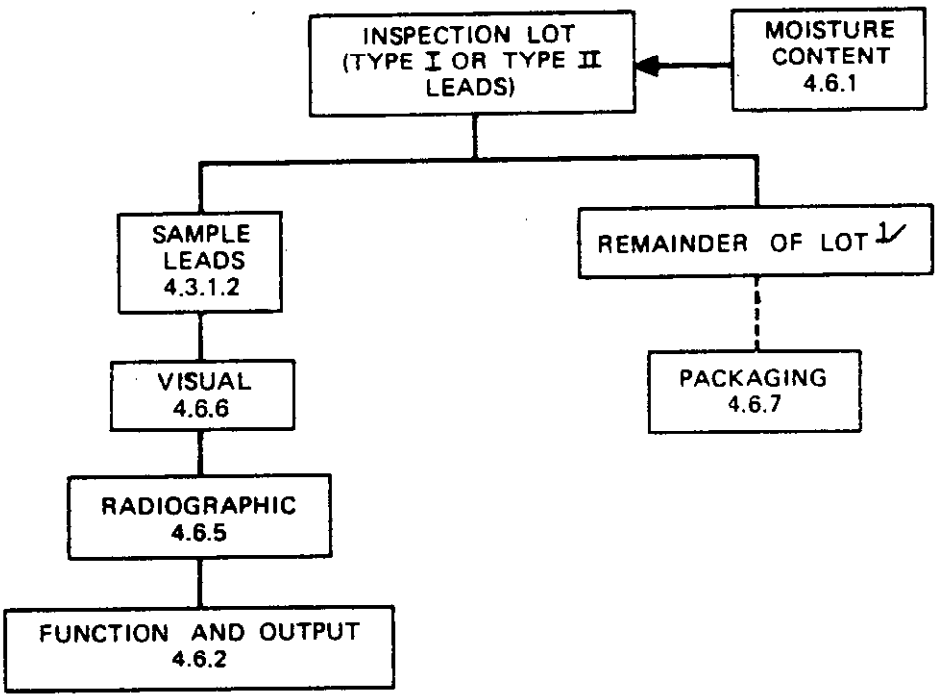


FIGURE 1. First article inspection sequence.

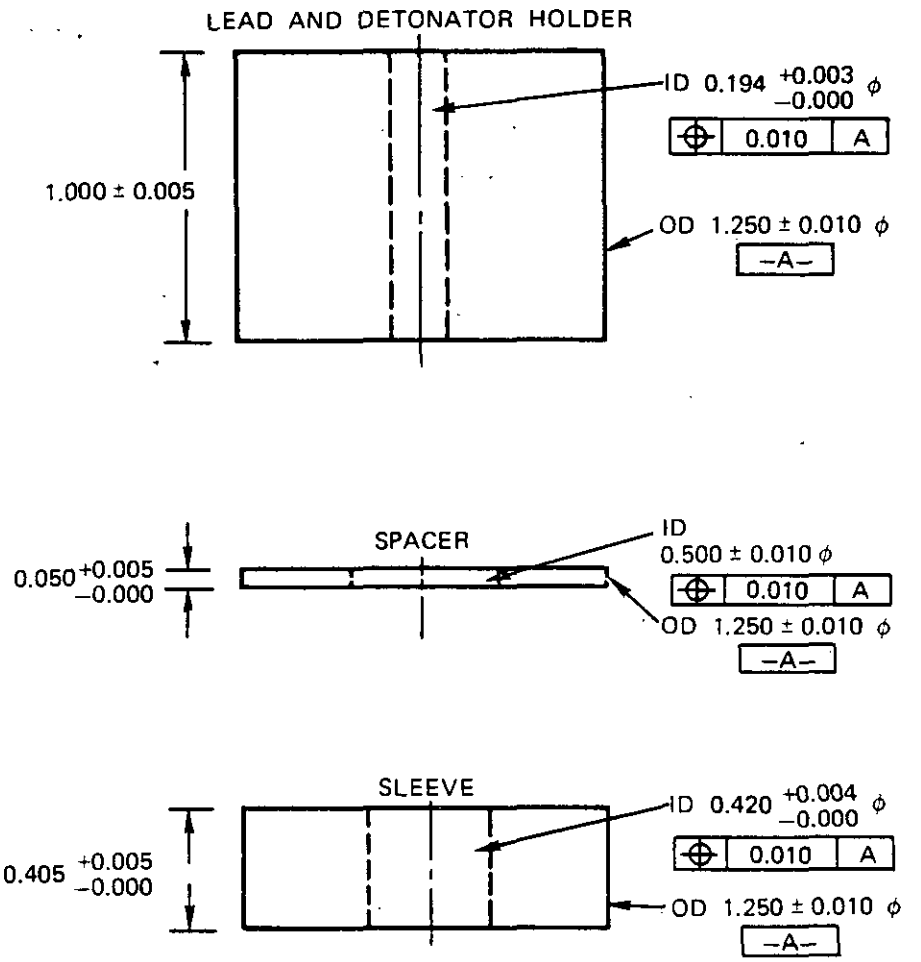
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1/ Remainder of lot shall be held in bond pending results of inspections on sample leads. If sample leads do not cause lot rejection, the remainder of the lot shall continue through the inspection sequence (see 4.3.1.2).

FIGURE 2. Inspection lot acceptance inspection sequence.

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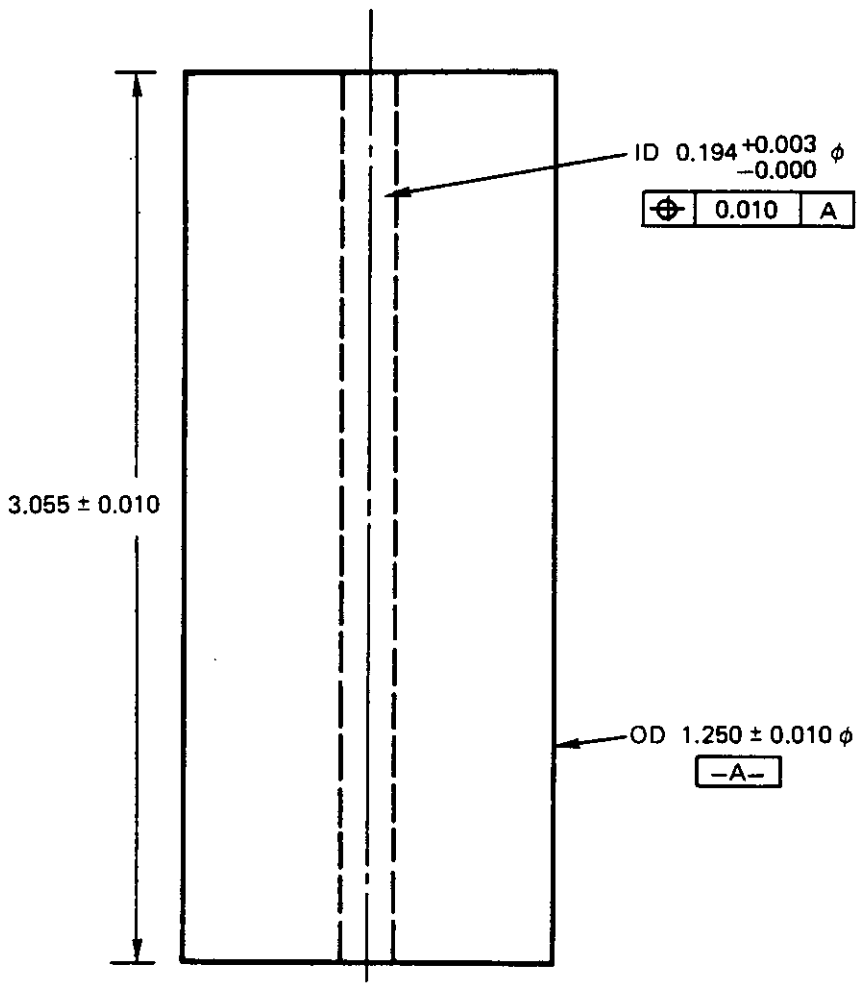


NOTES:

1. Material: Corrosion-resistant steel in accordance with ASTM A–582, Type 416, Condition A. For Class B steels, heat-treat in accordance with MIL–H–6875 to Rockwell hardness C25 to 35.
2. Dimensions are in inches.

FIGURE 3. Lead and detonator holder, spacer, and sleeve for function and output test of Type I lead.

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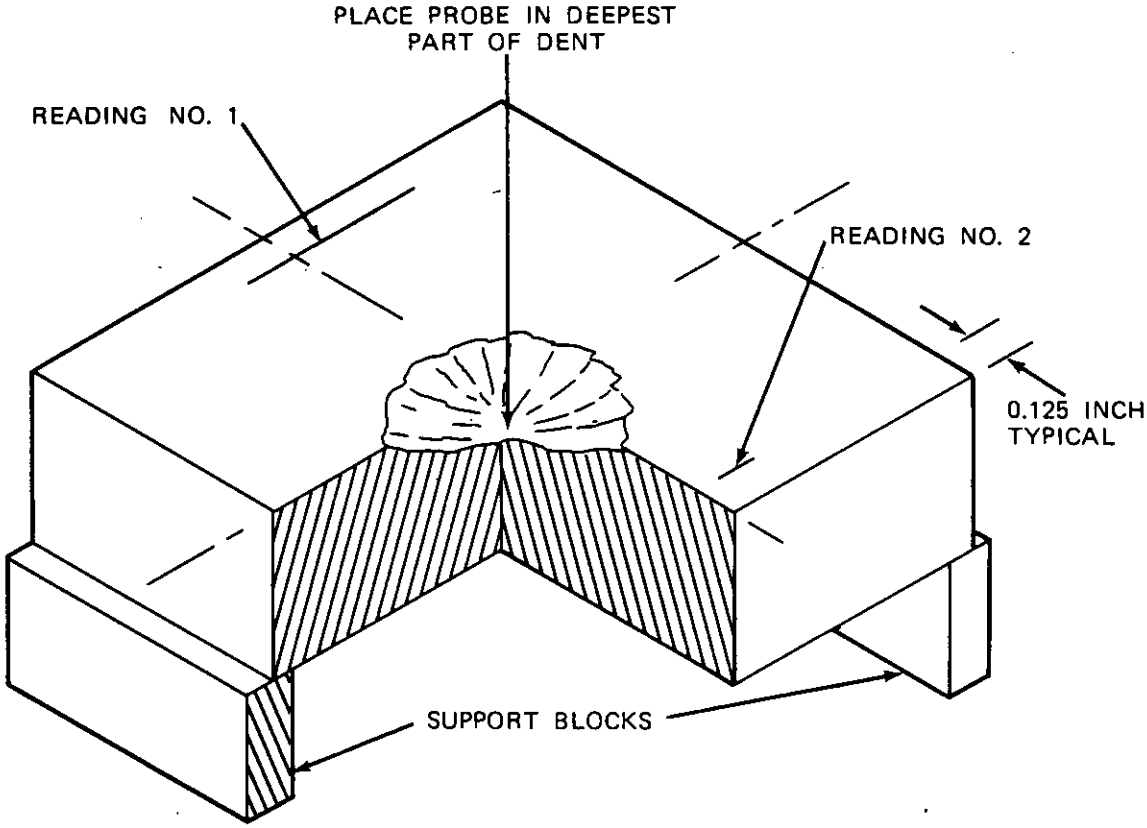


NOTES:

1. Material: Corrosion-resistant steel in accordance with ASTM A–582, Type 416, Condition A. For Class B steels, heat-treat in accordance with MIL–H–6875 to Rockwell hardness C25 to 35.
2. Dimensions are in inches.

FIGURE 4. Lead and detonator holder for function and output test of Type II lead.

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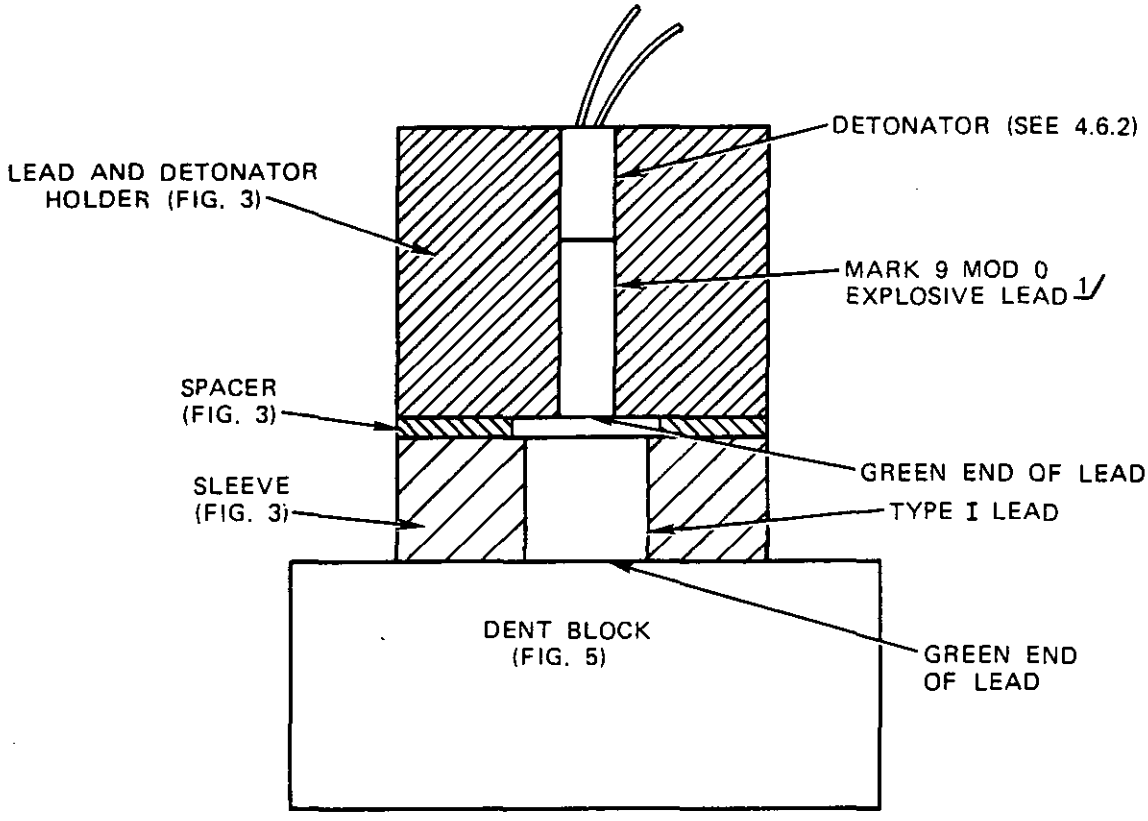


NOTES:

1. Dent Block: Steel, cold finished bar in accordance with ASTM A-108, Grade 1018 or 1020, Rockwell hardness B70 to B95, surface condition as rolled. Size 2.000 ±0.125 inches square, 1.5 inch stock. Upper and lower surfaces of block shall be clean and free from rust and burrs.
2. Support Blocks: Material and size shall be at the contractor's discretion.

FIGURE 5. Dent block and dent measurement locations for function and output test.

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1/ Using a suitable adhesive (type of adhesive shall be at the contractor's discretion), the Mark 9 Mod 0 Explosive Lead shall be secured in the lead and detonator holder such that the base of the lead is flush , ±0.010, with the bottom of the holder as shown.

FIGURE 6. Function and output test setup for Type I lead.

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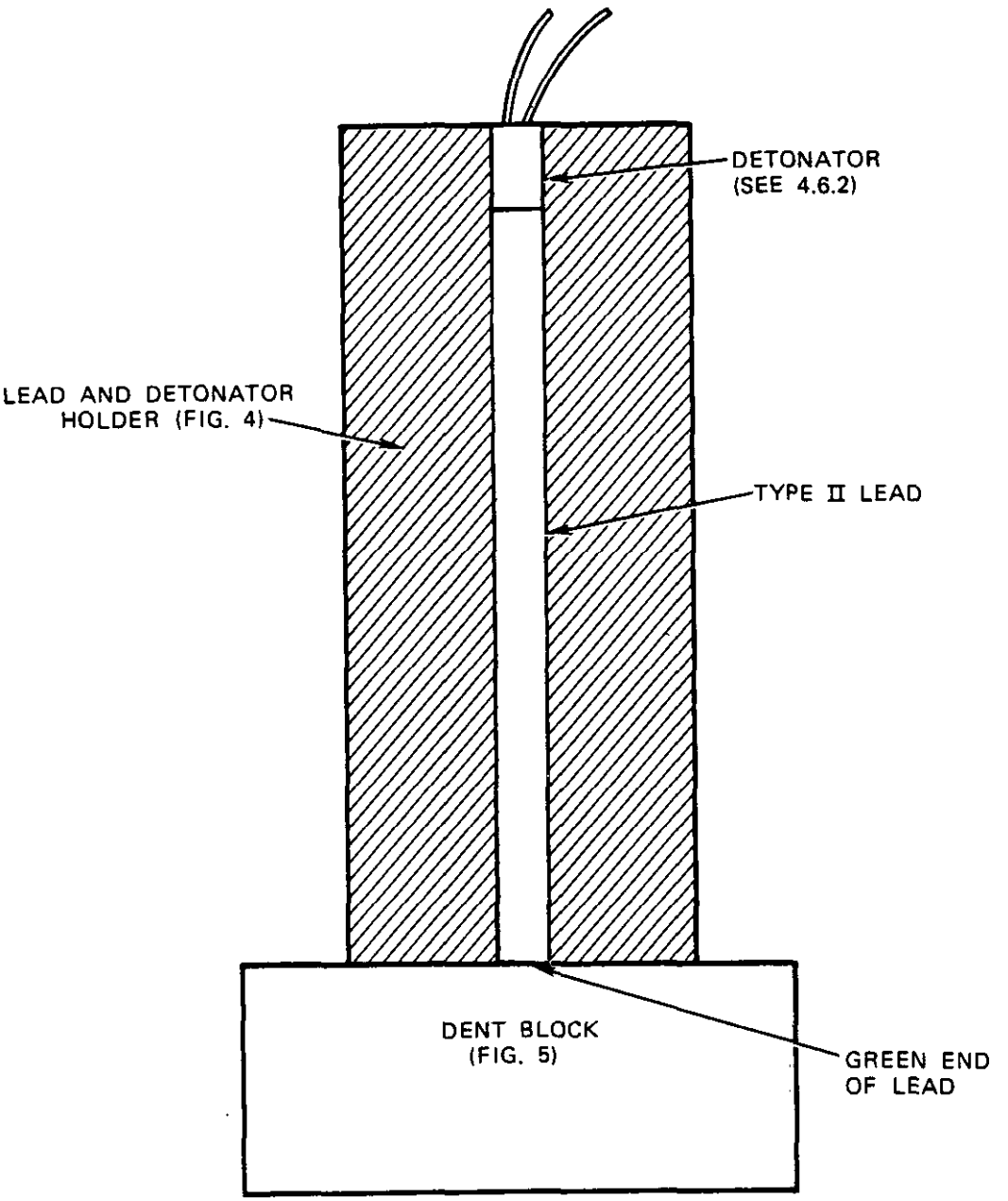
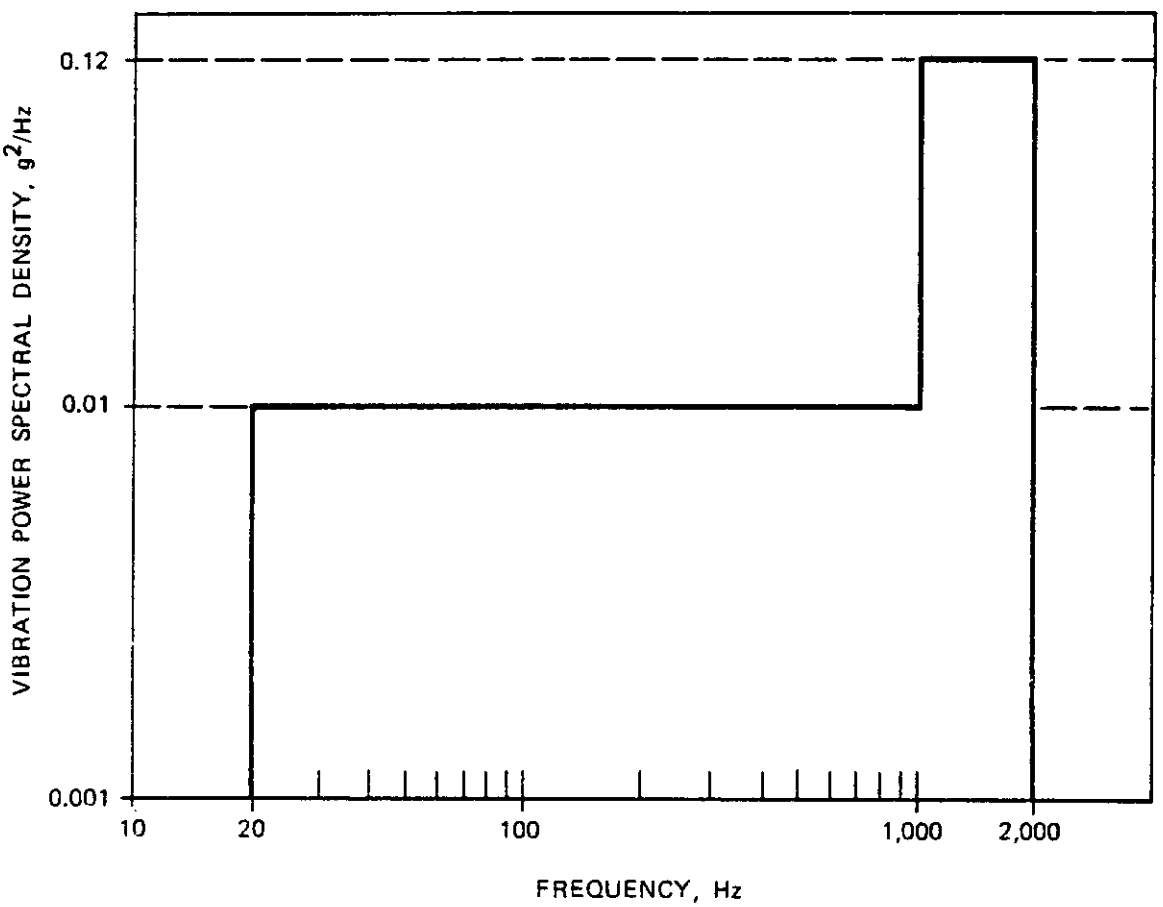


FIGURE 7. Function and output test setup for Type II lead.

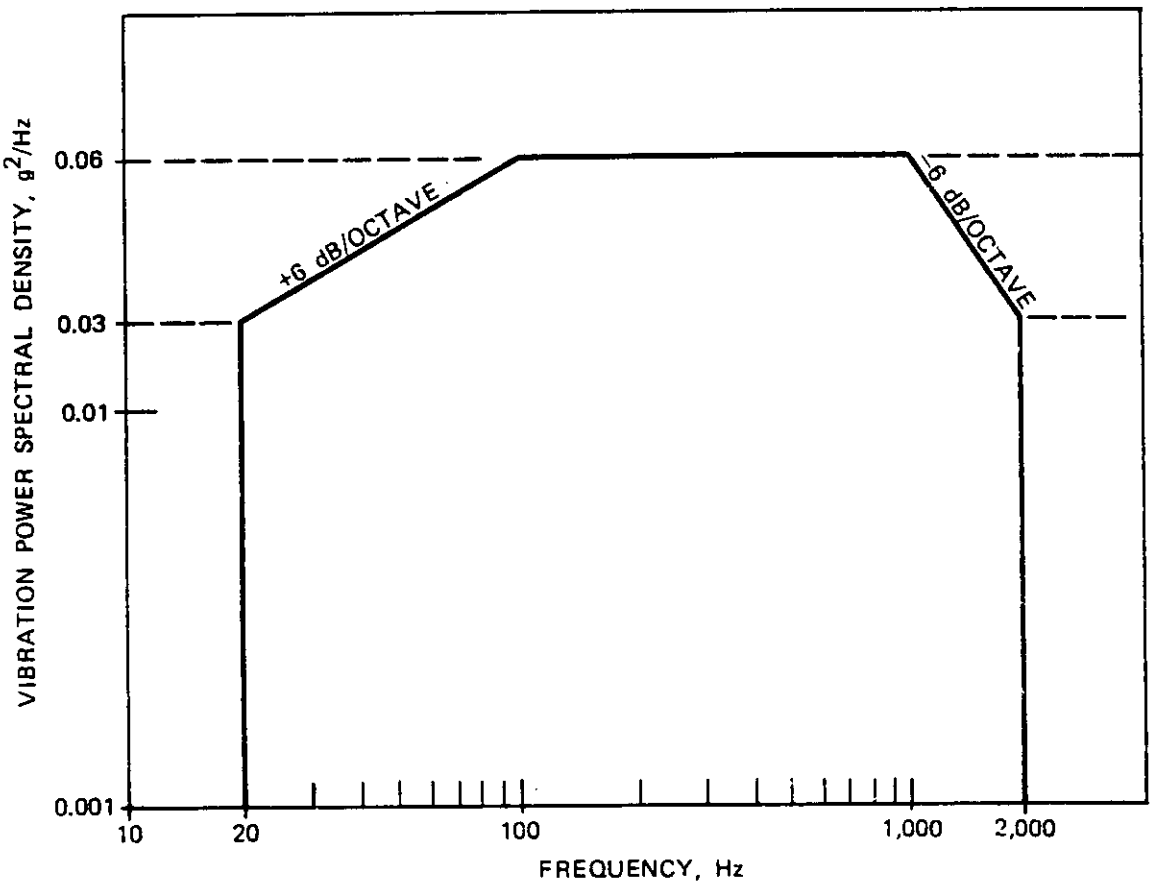
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NOTE: Vibration shall be applied along each of the three mutually perpendicular axes for 60 minutes minimum per axis and as specified in 4.6.3.3..

FIGURE 8. Captive-flight vibration spectrum.

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NOTE: Vibration shall be applied along each of the three mutally perpendicular axes for 5 minutes minimum per axis and as specified in 4.6.3.3

FIGURE 9. Free-flight vibration spectrum.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-85602(AS)	"Lead, Explosive, MARK 28 MOD 0, MARK 29 MOD 0"

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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